ECON6067 Individual Final Project

Cash Transfers and Local General Equilibrium

Replication of Egger et al. (2022) Table V, with a rollout-timing and exposure-network robustness extension

Student	Zhenyu Yang
Student ID	3036056641
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Repository	github.com/StupidKevin7743/ZhenyuYang_3036056641

APPROVAL STATUS: Egger et al. is an alternative paper outside the four illustrative options in the project brief. Instructor approval and target confirmation are requested in the accompanying unsent email draft; approval has not been assumed in this report.

Reproducibility statement: the repository contains code, disclosure-reviewed aggregate outputs, access instructions, an AI-use record, and one public entry point. It contains no microdata, household identifiers, coordinates, or restricted GPS inputs.

1. Executive summary

Egger et al. (2022) study a one-time cash-transfer program that paid about USD 1,000 nominal to more than 10,500 poor households in 653 randomized Kenyan villages. Because the transfer shock exceeded 15% of local GDP, the experiment can measure both direct recipient spending and indirect local spillovers. The paper's central result is a local transfer multiplier of roughly 2.5 with little aggregate price inflation.

This project targets Table V. It reproduces the published dynamic-IV point estimates and validates a matrix-based Python implementation against Stata to within 1.8e-7. It then performs 2,000 Rademacher wild-cluster draws at the sublocation level. The reimplemented expenditure multiplier is 2.5767 (SE 1.4575), the income multiplier is 2.4691 (SE 1.7874), and their average is 2.5229 (SE 1.4278). These closely track the printed estimates 2.58, 2.47, and 2.52.

The independent extension asks whether rollout timing and the nearby-treatment network create systematic selection or very thin comparison cells. Conditional on subcounty fixed effects, timing is not detectably related to village size, eligibility share, or saturation assignment. Yet at a 2 km radius there is only one direct-only village and four pure-control villages; those categories disappear at 3 km and 4 km. This is an identification diagnostic, not a new treatment-effect estimate.

What the project establishes

- Core Table V point estimates and presentation are reproduced.
- The completed 2,000-draw accelerator closely reconciles published uncertainty and p-values.
- The extension makes rollout phases and exposure-cell support auditable.

What the project does not establish

- It is not a full-paper, paper-identical end-to-end replication.
- It does not reproduce the restricted-coordinate spatial-HAC implementation.
- The direct Stata 2,000-repetition loop did not finish; the final inference is the validated Python accelerator.

2. Research question and economic mechanism

The paper asks whether a large, externally financed cash transfer raises only the recipient household's expenditure or also expands local production and income. A purely partial-equilibrium view predicts direct consumption gains. A local general-equilibrium view adds demand spillovers: recipients buy local goods and services, firms respond through sales, inventories, investment, profits, wages, and taxes, and nonrecipients receive part of the resulting income flow.

The multiplier divides the integrated local expenditure or income response by the transfer amount. Values greater than one are possible when one household's spending becomes another household's or firm's income and is spent again. Whether this occurs without large inflation depends on local supply elasticities, unused capacity, trade, and leakage outside the study area.

Why Table V

Table V aggregates the paper's dynamic expenditure and income responses into the headline multipliers and decomposes them into economically interpretable components. The paper's Figure 1 provides the quarterly-path context, but this course package does not claim to regenerate that exhibit. The selected table connects the research question, identification strategy, dynamics, and main conclusion more directly than a descriptive balance table would.

Predictions assessed

Prediction	Observable implication
Local demand propagates	Expenditure and income multipliers exceed zero and may exceed one.
Supply responds	Enterprise profits, investment, inventories, wages, and taxes rise.
Effects unfold over time	Impulse responses persist across quarterly lags rather than appearing in one period.
Design support matters	Rollout and exposure cells provide credible variation for own and nearby transfers.

The extension focuses on the fourth prediction. It does not redefine the paper's estimand; it tests whether the timing and simplified exposure geometry reveal support problems that should qualify interpretation.

3. Data, sample, and experimental design

The experiment covers 653 villages in 84 sublocations: 328 treated and 325 control. Randomization is hierarchical. Sublocations were assigned to high- or low-saturation groups, and villages within them were treated at different probabilities. Eligible households in treated villages received KES 87,000 in three payments: KES 7,000, then KES 40,000 about two months later, then KES 40,000 about six months later.

The household census contains 65,383 households (about 280,000 people). Baseline surveys cover 7,845 households and endline surveys 8,239, with tracking above 90%. Endline occurred a median 19 months after experimental start and about 11 months after the final transfer. Enterprise and market data supply the remaining multiplier components.

Inputs used by this package

Logical input	Role	Public-repository treatment
HH/enterprise multiplier analysis file	Dynamic-IV components for Table V	Access instructions only; never committed
Stata multiplier estimate file	Point-estimate cross-check	Generated/authorized local input; never committed
Experimental timing panel	Actual rollout month	Access instructions only; never committed
Household/village analysis files	Village size, eligibility, treatment	Access instructions only; never committed
Public village-buffer file	Counts of nearby treated villages	No coordinates committed; input remains untracked

Disclosure boundary

Only aggregate CSV tables and two figures are versioned. The repository's release auditor uses an exact output allowlist and rejects Stata data, coordinates, archives, logs, identifiers, machine-local paths, unexpected result files, and credentials. Legacy public execution fixes **runGPS = 0**; no claim is made about restricted GPS analysis.

4. Identification and replication method

The paper's dynamic IV specification relates quarterly outcomes to own-village and nearby-village cumulative transfer amounts within 0-2 km across ten quarterly lags. Random assignment of village treatment, saturation, and rollout timing instruments for realized transfer exposure. Flow responses are integrated over 29 months; stock outcomes are measured at endline. Values are expressed in January 2015 terms.

Core workflow

Stage	Implementation	Validation
1. Point estimates	Reconstruct the Table V dynamic-IV component vector	Compare 11 quantities with Stata
2. Resampling	2,000 Rademacher wild-cluster draws by sublocation	Check exact draw count and fixed seed
3. Aggregation	Form expenditure, income, average, and joint tests	Compare printed precision with Table V
4. Presentation	Write aggregate CSVs and render figures	Schema/privacy audit plus visual review

Inference

For each wild-bootstrap draw, sublocation-level Rademacher weights perturb the null residual structure. The empirical distribution yields bootstrap standard errors and one-sided probabilities Pr(M < 0) and Pr(M < 1). This matches the paper's stated 2,000 within-sublocation replication logic. The Python code is a matrix accelerator, not a claim that the original Stata loop completed.

Pre-specified comparison rule

A point estimate passes if the Python-Stata absolute difference is below 1e-6. A printed benchmark is treated as reconciled when the reimplemented value agrees at the paper's reported precision or its small deviation is reported explicitly. No manual edits are made to canonical output numbers.

Why not use the earlier three-draw result?

The earlier Stata smoke run reproduces point estimates but produces p-values only in increments of one third and unstable standard errors. It is retained as pipeline evidence only and excluded from the canonical course outputs.

5. Core replication results

The core point estimates match the paper at printed precision. The expenditure multiplier is 2.5767; the income multiplier is 2.4691; their average is 2.5229. The 2,000-draw standard errors are slightly larger than the published values by 0.018, 0.077, and 0.038, respectively. The reimplemented one-sided p-values remain close to those printed in Table V.

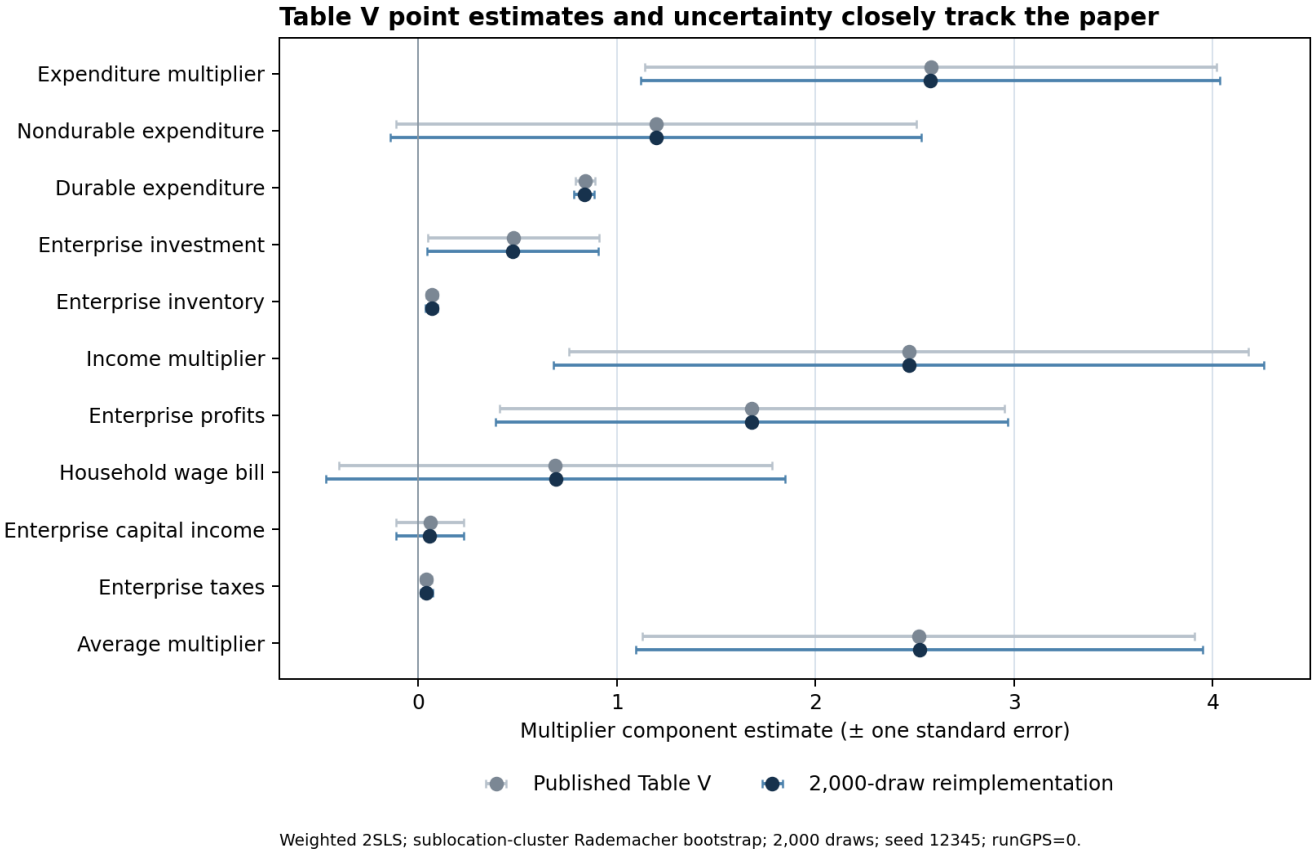
Component	Paper	Reimpl.	Paper SE	Reimpl. SE	p<0	p<1
Expenditure multiplier	2.58	2.58	1.44	1.46	0.030	0.145
Household non-durable expenditure	1.20	1.20	1.31	1.34	0.191	-
Household durable expenditure	0.84	0.84	0.05	0.05	0.000	-
Enterprise investment	0.48	0.48	0.43	0.43	0.143	-
Enterprise inventory	0.07	0.07	0.03	0.03	0.011	-
Income multiplier	2.47	2.47	1.71	1.79	0.086	0.210
Enterprise profits	1.68	1.68	1.27	1.29	0.103	-
Household wage bill	0.69	0.69	1.09	1.15	0.274	-
Enterprise capital income	0.06	0.06	0.17	0.17	0.363	-
Enterprise taxes paid	0.04	0.04	0.03	0.03	0.096	-
Average of both multipliers	2.52	2.52	1.39	1.43	0.036	0.149

Table 1. Published versus reimplemented Table V. p<0 and p<1 are the reimplemented one-sided wild-bootstrap probabilities. A dash means the paper did not report that component-level test.

Interpretation

The expenditure multiplier is significantly above zero at conventional one-sided levels but not significantly above one (Pr[M<1] = 0.1455). Income has Pr(M<0) = 0.0860 and Pr(M<1) = 0.2105. The joint probabilities for expenditure and income are 0.0130 against zero and 0.0785 against one. The evidence supports a positive local response; it is more cautious about claiming a multiplier definitively above one.

6. Visual comparison and numerical validation



Report Figure 1. Published and reimplemented Table V point estimates, with plus/minus one standard error. The two series nearly overlap. Error bars are descriptive rather than 95% intervals.

Point-estimate cross-check

Across 11 Stata quantities, the largest absolute Python-Stata difference is 1.7844e-7, below the 1e-6 tolerance. This isolates uncertainty differences from coefficient-construction errors.

Reconciliation, not overclaiming

The reimplementation does not reproduce every printed digit of every standard error. Potential sources include numerical details of the original Stata routines, null imposition, and random-number implementation. The magnitudes and inference closely track the published benchmark, but the result is labeled an accelerated reimplementation rather than a completed original-code bootstrap.

7. Independent extension: rollout and exposure support

The extension asks: **Do rollout timing and nearby-treatment exposure create systematic selection or thin comparison cells that qualify the interpretation of general-equilibrium spillovers?** It has two parts. First, actual start month is regressed on pre-existing village scale, eligibility share, and high-saturation assignment, with subcounty fixed effects and sublocation-clustered standard errors. Second, villages are classified as direct only, direct plus indirect, pure control, or spillover only at 2, 3, and 4 km.

Timing model	Covariate	Coef.	SE	p-value
First token	log weighted village households	-0.064	0.205	0.755
First token	eligible share	-0.207	0.995	0.835
First token	high saturation	0.168	0.286	0.557
10% cutoff	log weighted village households	-0.047	0.202	0.818
10% cutoff	eligible share	-0.084	1.017	0.934
10% cutoff	high saturation	0.185	0.287	0.519

Table 2. Timing diagnostics, N = 328 treated villages and 80 sublocation clusters. Subcounty fixed effects are included but omitted from the display.

Result 1: no detected within-subcounty timing gradient

The coefficients on log weighted households, eligible share, and high saturation are individually small relative to their standard errors in both timing definitions. The subcounty fixed effects are large, consistent with phased geographic rollout. These are descriptive correlations and do not convert rollout timing into an outcome estimate.

Result 2: binary exposure comparisons become unsupported

At 2 km there are 1 direct-only, 327 direct-plus-indirect, 4 pure-control, and 321 spillover-only villages. At 3 km and 4 km, the direct-only and pure-control cells vanish. Hence a simple four-cell comparison would be unstable; the paper's continuous exposure design is more appropriate.

8. Extension evidence and implications

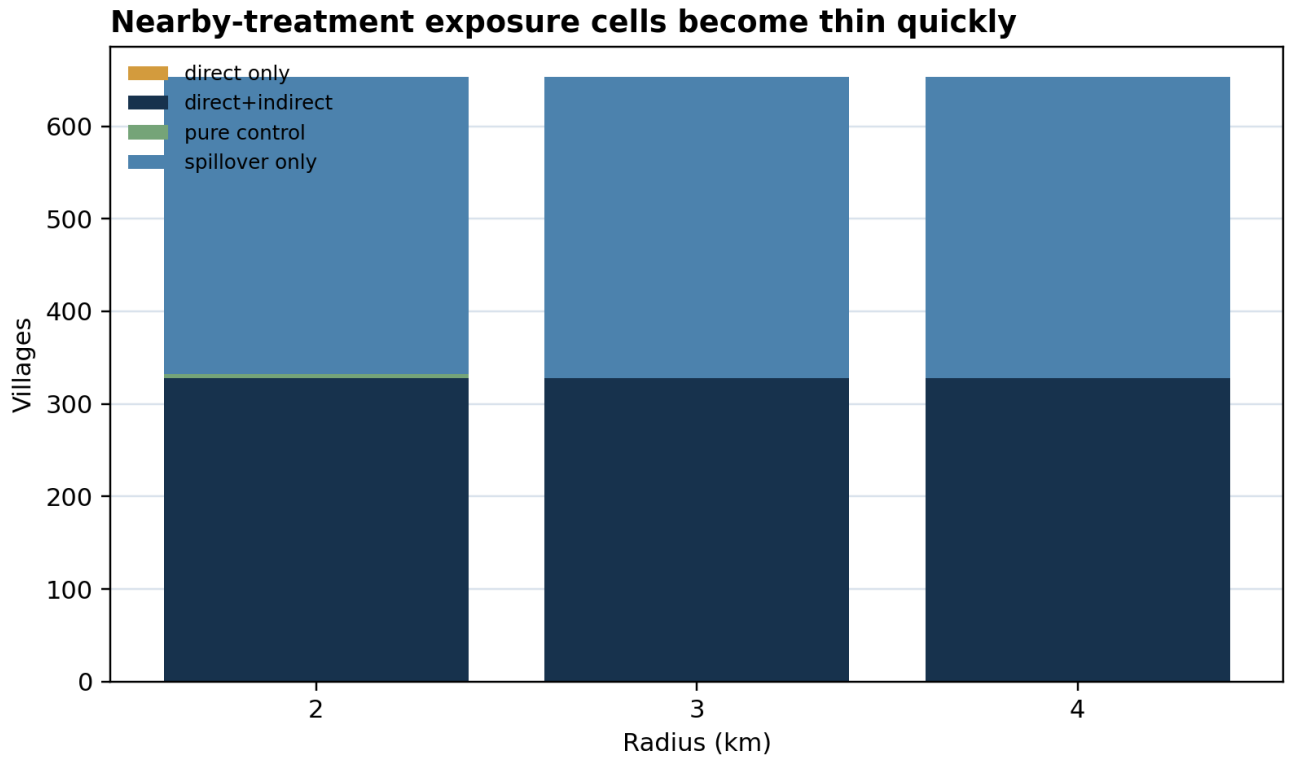


Figure 2. Simplified village exposure cells at 2, 3, and 4 km. Each village appears once within a treated or control bar component for each radius.

Economic and design interpretation

Within 2 km, villages have on average 13.88 neighbors (median 14; maximum 26). Two villages have no neighbors and five have no treated neighbor. Direct-only and pure-control villages represent only 0.3% and 0.9% of weighted households. That lack of support is not evidence against spillovers; it is evidence that binary cell means would be a poor substitute for continuous nearby-transfer exposure.

The extension therefore strengthens interpretation in a limited but useful way. Observed timing does not show an obvious within-subcounty association with the three tested village characteristics, while the geometry explains why instruments based on randomized saturation and continuous exposure are needed. It also warns against claiming clean direct-only versus pure-control contrasts at broader radii.

Extension limitations

- Monthly timing cannot recover exact weekly payment order.
- The exposure cells reduce a continuous dose to any-versus-none categories.
- Timing regressions cover three observed characteristics, not every source of selection.
- No new causal outcome coefficient is estimated in this extension.

9. Reproducibility package

The public repository preserves the upstream Stata/R workflow and adds a course-facing Python entry point. From an authorized local copy of the documented inputs, the intended command is **python3 code/run_all.py --target all**. Explicit path overrides support inputs stored outside the repository. A check-only mode validates the committed package without requiring restricted data.

Package map

Path	Purpose
code/run_all.py	Single course entry point; preflight, core, extension, output validation
code/table5_multiplier_bootstrap.py	Portable Table V point estimates and 2,000-draw accelerator
code/extension_timing_exposure.py	Timing regression and exposure-support diagnostics
code/render_figures.py	Figures generated only from aggregate CSV outputs
config/	Project status, input contract, and published benchmarks
data/raw/ and data/processed/	README access instructions only; inputs remain untracked
outputs/tables/ and outputs/figures/	Disclosure-reviewed canonical aggregate artifacts
skills/SKILL.md	Reusable execution, checks, privacy invariants, and troubleshooting
AI_USE_DISCLOSURE.md	Stage-by-stage AI assistance and human verification

Fresh-run controls

- Resolve paths from the repository root; no machine-local path is committed.
- Fail before analysis if inputs, required variables, or output schemas are missing.
- Use a fixed seed and 2,000 draws in paper mode; smoke results cannot replace canonical outputs.
- Reject GPS/coordinate files and keep legacy public analysis at runGPS = 0.
- Run the exact-path privacy auditor before any push.

The committed results were regenerated from the existing authorized local analysis files and compared with the paper. A third party still needs lawful access to the documented inputs to reproduce the numerical run from scratch.

10. Validation, limitations, and AI use

Validation completed

- Exact paper and appendix identities were verified from contents; key exhibits were rendered.
- The Table V summary contains exactly 2,000 draw rows in its supporting draw file.
- All 11 Python point estimates pass the 1e-6 Stata-difference tolerance.
- Extension tables were freshly regenerated and match prior aggregate results.
- Canonical CSV schemas, figure dimensions, PDF extraction, and rendered pages were checked.

Substantive limitations

- Alternative-paper approval remains pending; this is a submission candidate, not evidence of approval.
- The package targets Table V plus one robustness extension, not all paper exhibits or the paper's Figure 1.
- Restricted spatial-HAC/GPS inference is not reproduced in the public package.
- A direct 2,000-draw Stata loop was incomplete; the final results use the validated Python matrix implementation.
- Early quarterly responses are imprecise because survey timing begins well after rollout for some outcomes.

AI use and human responsibility

AI assisted with requirement extraction, repository auditing, portability edits, table/figure generation, report drafting, and validation planning. Every empirical claim in this report was checked against the named paper, appendix, code, logs, or aggregate output. Suggestions that would have mislabeled three-draw inference, claimed a complete Stata run, or treated a third-party reanalysis as the student's own extension were rejected. Full details appear in AI_USE_DISCLOSURE.md.

11. Conclusion

The selected replication supports the paper's headline result in a carefully bounded sense. Table V point estimates are reproduced to numerical precision, and a validated 2,000-draw wild-cluster accelerator closely reconciles the reported uncertainty. The expenditure multiplier is about 2.58, the income multiplier about 2.47, and their average about 2.52. The evidence is strong that the local response is positive; the one-sided evidence that the multiplier exceeds one is suggestive rather than decisive.

The rollout/exposure extension adds a design insight. Within subcounty, start timing is not detectably associated with the tested village characteristics, but simplified direct-only and pure-control exposure cells are extremely thin and disappear at broader radii. This supports the paper's use of randomized saturation and continuous nearby transfer exposure while warning against unsupported binary comparisons.

The main remaining decision is administrative rather than computational: instructor approval is required because Egger et al. is an alternative paper. Subject to that approval, the repository now matches the requested submission structure and states its replication boundary honestly.

References

Egger, D., Haushofer, J., Miguel, E., Niehaus, P., and Walker, M. (2022). General Equilibrium Effects of Cash Transfers: Experimental Evidence From Kenya. *Econometrica*, 90(6), 2603-2643. https://doi.org/10.3982/ECTA17945.

Egger et al. Complete Appendices A-J (version dated 7 June 2022), especially Appendix A (design), Appendix C (MPC), and Appendix D (multiplier robustness).

Rubric map

Project criterion	Report location
Understanding and motivation	Sections 1-2
Data sources, sample, variables	Section 3
Core replication and validation	Sections 4-6
Independent extension	Sections 7-8
Reproducibility/workflow	Section 9 and repository
Limitations, AI disclosure, conclusion	Sections 10-11 and AI_USE_DISCLOSURE.md